

Architecture Model

Project: architecture-model-standard **Schema Version:** 1.4

Actors

ACT-DEV: Developer [ACTIVE]

- **description:** Software engineer using CLI to manage architecture models

ACT-LLM: LLM Agent [ACTIVE]

- **description:** AI agent consuming architecture models via opencode-arch MCP tools

Layers

L-APP: Application Layer [ACTIVE]

Capabilities

CAP-F1: Model Parsing & Validation [ACTIVE]

- **f_block:** F3
- **description:** Parse YAML models and validate structural correctness (score 0-100)

CAP-F2: Reality Manifest Generation [ACTIVE]

- **f_block:** F4
- **description:** AST scanning to produce ground-truth code inventories with metrics, body hints, and test contracts

CAP-F3: Model Slicing & Diffing [ACTIVE]

- **f_block:** F3
- **description:** Subset extraction by F-block/layer, model version comparison, and enriched model composition

CAP-F4: CLI Operations [ACTIVE]

- **f_block:** F1
- **description:** Command-line interface for init, validate, slice, diff, stats, impact, manifest

CAP-F5: Configuration Management [ACTIVE]

- **f_block:** F2
- **description:** Auto-discovery config loading and project configuration schema

CAP-F6: Schema Specification [ACTIVE]

- **f_block:** F5
- **description:** JSON Schema definitions for architecture model validation

Behaviors

BEH-INIT: Project Initialization [ACTIVE]

- **description:** Scan directory structure, discover source root, auto-generate .architecture-model.yaml

BEH-VALIDATE: Model Validation [ACTIVE]

- **description:** Check architectural claims against schema and structural invariants, produce score 0-100

BEH-MANIFEST: Manifest Generation [ACTIVE]

- **description:** Walk source tree, parse ASTs, extract functions/classes/imports, compute metrics, discover blocks

Interfaces

IF-CLI: CLI Interface [ACTIVE]

- **description:** Entry point: architecture-model {init,validate,slice,diff,stats,impact,manifest}

IF-PARSE-API: Parser API [ACTIVE]

- **description:** load_model(path) -> ArchitectureModel, save_model(model, path), dump_model(model) -> str

IF-VALIDATE-API: Validator API [ACTIVE]

- **description:** validate_model(model) -> ValidationResult (score, issues, is_valid)

IF-MANIFEST-API: Manifest API [ACTIVE]

- **description:** generate_manifest(project_root) -> dict with modules, metrics, functional_blocks, interfaces

IF-SLICER-API: Slicer API [ACTIVE]

- **description:** slice_by_fblock(model, id), slice_by_layer(model, id), slice_for_artifact(model, artifact)

Constraints

CON-SCHEMA: Schema Compliance [ACTIVE]

- **description:** All models must conform to JSON Schema v1.4 with 7 entity types and 8 relationship types

CON-NO-ORPHANS: No Orphaned Entities [ACTIVE]

- **description:** Every entity must participate in at least one relationship

Components

COMP-CORE: core [ACTIVE]

- **description:** Parser, validator, slicer, differ, merger, decomposer, type system (ArchitectureModel, Component, Capability, etc.)

COMP-MANIFEST: manifest [ACTIVE]

- **description:** AST scanner, block discovery, metrics, interface detection, body hints, test analyzer, display, slicers

COMP-CONFIG: config [ACTIVE]

- **description:** Configuration loader (auto-discovery) and schema definition

COMP-SPEC: spec [ACTIVE]

- **description:** JSON Schema definitions for architecture model validation

COMP-CLI: cli [ACTIVE]

- **description:** CLI entry point and command dispatch (7 commands: init, validate, slice, diff, stats, impact, manifest)

Relationships

Type	From	To
contains	L-APP	COMP-CORE
contains	L-APP	COMP-MANIFEST
contains	L-APP	COMP-CLI
contains	L-APP	COMP-CONFIG
contains	L-APP	COMP-SPEC
realizes	COMP-CORE	CAP-F1
realizes	COMP-CORE	CAP-F3
realizes	COMP-MANIFEST	CAP-F2
realizes	COMP-CLI	CAP-F4
realizes	COMP-CONFIG	CAP-F5
realizes	COMP-SPEC	CAP-F6
exposes	COMP-CLI	IF-CLI
exposes	COMP-CORE	IF-PARSE-API
exposes	COMP-CORE	IF-VALIDATE-API
exposes	COMP-CORE	IF-SLICER-API
exposes	COMP-MANIFEST	IF-MANIFEST-API
constrained-by	COMP-CORE	CON-SCHEMA
constrained-by	COMP-SPEC	CON-SCHEMA
constrained-by	COMP-CORE	CON-NO-ORPHANS
depends-on	COMP-CLI	COMP-CORE
depends-on	COMP-CLI	COMP-CONFIG
depends-on	COMP-CLI	COMP-MANIFEST
depends-on	COMP-CORE	COMP-CONFIG
depends-on	COMP-CORE	COMP-SPEC
depends-on	COMP-MANIFEST	COMP-CONFIG

Type	From	To
traces-to	COMP-CLI	BEH-INIT
traces-to	COMP-CORE	BEH-VALIDATE
traces-to	COMP-MANIFEST	BEH-MANIFEST
consumes	ACT-DEV	IF-CLI
consumes	ACT-LLM	IF-PARSE-API
consumes	ACT-LLM	IF-SLICER-API